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3rd Edition

Data Science

for dummies®

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Deploy math and stats to improve decision-making

Supercharge your business growth with data science wins

Monetize your data resources and data science skills

Lillian Pierson, PE

CEO, Data-Mania



Data Science

3rd Edition

by Lillian Pierson

for dummies®
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Introduction

This book was written as much for expert data scientists as it was for aspiring ones. Its content represents a new approach to doing data science — one that puts business vision and profitably at the heart of our work as data scientists.

Data science and artificial intelligence (AI, for short) have disrupted the business world so radically that it's nearly unrecognizable compared to what things were like just 10 or 15 years ago. The good news is that most of these changes have made everyone’s lives and businesses more efficient, more fun, and dramatically more interesting. The bad news is that if you don’t yet have at least a modicum of data science competence, your business and employment prospects are growing dimmer by the moment.

Since 2014, when this book was first written (throughout the first two editions), I have harped on this same point. Lots of people listened! So much has changed about data science over the years, however, that this book has needed two full rewrites since it was originally published. What changed? Well, to be honest, the math and scientific approach that underlie data science haven’t changed one bit. But over the years, with all the expansion of AI adoption across business and with the remarkable increase in the supply of data science workers, the data science landscape has seen a hundredfold increase in diversity with respect to what people and businesses are using data science to achieve.

The original idea behind this book when it was first published was to provide “a reference manual to guide you through the vast and expansive areas encompassed

by data science.” At the time, not too much information out there covered the breadth of data science in one resource. That has changed!

Data scientist as a title only really began to emerge in 2012. Most of us practitioners in the field back then were all new and still finding our way. In 2014, I didn’t have the perspective or confidence I needed to write a book like the one you're holding. Thank you so much to all the readers who have read this book previously, shared positive feedback, and applied what they learned to create better lives for themselves and better outcomes for their companies. The positive transformation of my readers is a big part of what keeps me digging deep to produce the very best version of this book that I possibly can.

The Internet is full of *information for the sake of information* — information that lacks the depth, context, and relevance that's needed to transform that information to true meaning in the lives of its consumers. Publishing more of this type of content doesn’t help people — it confuses them, overwhelms them, and wastes their precious time! When writing this book for a third time, I took a radical stance against “information for the sake of information.”

I also want to make three further promises about the content in this book: It is meaningful, it is actionable, and it is relevant. If it isn’t one of these three adjectives, I’ve made sure it hasn’t made its way into this book.

Because this book is about data science, I spend the entirety of Parts 1 and 2 detailing what data science actually is and what its theoretical underpinnings are. Part 3 demonstrates the ways you can apply data science to support vital business functions, from finance to marketing, from decision support to operations. I’ve even

written a chapter on how to use data science to create what may be a whole new function within your company: data monetization. (To ensure that the book's content is relevant to readers from all business functions and industries, I've included use cases and case studies from businesses a wide variety of industries and sectors.)

To enhance the relevance of this book's content, at the beginning of the book I guide readers in a self-assessment designed to help them identify which type of data science work is most appropriate for their personality — whether it's implementing data science, working in a management and leadership capacity, or even starting your own data science business.

Part 4 is the actionable part of this book — the part that shows you how to take what you've learned about data science and apply it to start getting results right away. The action you learn to take in this book involves using what you learn about data science in Parts 1 through 3 to build an implementation plan for a profit-forming data science project.

Throughout this book, you'll find references to ancillary materials that directly support what you're learning within these pages. All of these support materials are hosted on the website that companions this book, http://businessgrowth.ai/. I highly recommend you take advantage of those assets, as I have donated many of them from an archived, limited-edition, paid product that was only available in 2020.

Note: I have removed all coding examples from this book because I don't have adequate space here to do anything meaningful with coding demos. If you want me to show you how to implement the data science that's discussed in Part 2, I have two Python for Data Science Essential Training courses on LinkedIn Learning. You're most

welcome to follow up by taking those courses. You access them both directly through my course author page on LinkedIn Learning here:
www.linkedin.com/learning/instructors/lillian-pierson-p-e

This book is unlike any other data science book or course on the market. How do I know? Because I created it from scratch based on my own unique experience and perspective. That perspective is based on almost 15 years of technical consulting experience, almost 10 of which have been spent working as a consultant, entrepreneur, and mentor in the data science space. This book is not a remake of what some other expert wrote in *their* book — it's an original work of art and a labor of love for me. If you enjoy the contents of this book, please reach out to me at lillian@data-mania.com and let me know. Also, for free weekly video training on data science, data leadership, and data business-building, be sure to visit and subscribe to my YouTube channel:
https://www.youtube.com/c/LillianPierson_Data_Business

Helping readers like you is my mission in life!

About This Book

In keeping with the *For Dummies* brand, this book is organized in a modular, easy-to-access format that allows you to use the book as an owner's manual. The book's chapters are structured to walk you through a clear process, so it's best to read them in order. You don't absolutely have to read the book through, from cover to cover, however. You can glean a great deal from jumping around, although now and then you might miss some important context by doing so. If you're already working in the data science space, you can skip the basic-level details about what data science is within Part 2 — but do read the rest of the book, because it's designed to

present new and immensely valuable knowledge for data science practitioners of all skill levels (including experts).

Web addresses appear in monofont. If you're reading a digital version of this book on a device connected to the Internet, you can click a web address to visit that website, like this: www.dummies.com.

Foolish Assumptions

In writing this book, I've assumed that readers are comfortable with advanced tasks in Microsoft Excel — pivot tables, grouping, sorting, plotting, and the like. Having strong skills in algebra, basic statistics, or even business calculus helps as well. Foolish or not, it's my high hope that all readers have subject matter expertise to which they can apply the skills presented in this book. Because data scientists need to know the implications and applications of the data insights they derive, subject matter expertise is a major requirement for data science.

Icons Used in This Book

As you make your way through this book, you see the following icons in the margins:



The Tip icon marks tips (duh!) and shortcuts you can use to make subject mastery easier.



REMEMBER Remember icons mark information that's especially important to know. To siphon off the most important information in each chapter, just skim the material represented by these icons.



TECHNICAL STUFF The Technical Stuff icon marks information of a highly technical nature that you can normally skip.



WARNING The Warning icon tells you to watch out! It marks important information that may save you headaches.

Beyond the Book

Data Science For Dummies, 3rd Edition, comes with a handy Cheat Sheet that lists helpful shortcuts as well as abbreviated definitions for essential processes and concepts described in the book. You can use this feature as a quick-and-easy reference when doing data science. To download the Cheat Sheet, simply go to www.dummies.com and search for *data science for dummies cheat sheet* in the Search box.

Where to Go from Here

If you're new to data science, you're best off starting from Chapter 1 and reading the book from beginning to end. If you already know the data science basics, I suggest that you read the last part of Chapter 1, skim Chapter 2, and then dig deep into all of Parts 3 and 4.

Part 1

Getting Started with Data Science

IN THIS PART ...

Get introduced to the field of data science.

Delve into vital data engineering details.

Discover your inner data superhero archetype.

Chapter 1

Wrapping Your Head Around Data Science

IN THIS CHAPTER

- **Deploying data science methods across various industries**
- **Piecing together the core data science components**
- **Identifying viable data science solutions to business challenges**
- **Exploring data science career alternatives**

For over a decade now, *everyone* has been absolutely deluged by data. It's coming from every computer, every mobile device, every camera, and every imaginable sensor — and now it's even coming from watches and other wearable technologies. Data is generated in every social media interaction we humans make, every file we save, every picture we take, and every query we submit; data is even generated when we do something as simple as ask a favorite search engine for directions to the closest ice cream shop.

Although data immersion is nothing new, you may have noticed that the phenomenon is accelerating. Lakes, puddles, and rivers of data have turned to floods and veritable tsunamis of structured, semistructured, and unstructured data that's streaming from almost every activity that takes place in both the digital and physical

worlds. It's just an unavoidable fact of life within the information age.

If you're anything like I was, you may have wondered, "What's the point of all this data? Why use valuable resources to generate and collect it?" Although even just two decades ago, no one was in a position to make much use of most of the data that's generated, the tides today have definitely turned. Specialists known as *data engineers* are constantly finding innovative and powerful new ways to capture, collate, and condense unimaginably massive volumes of data, and other specialists, known as *data scientists,* are leading change by deriving valuable and actionable insights from that data.

In its truest form, data science represents the optimization of processes and resources. Data science produces *data insights* — actionable, data-informed conclusions or predictions that you can use to understand and improve your business, your investments, your health, and even your lifestyle and social life. Using data science insights is like being able to see in the dark. For any goal or pursuit you can imagine, you can find data science methods to help you predict the most direct route from where you are to where you want to be — and to anticipate every pothole in the road between both places.

Seeing Who Can Make Use of Data Science

The terms *data science* and *data engineering* are often misused and confused, so let me start off by clarifying that these two fields are, in fact, separate and distinct domains of expertise. *Data science* is the computational

science of extracting meaningful insights from raw data and then effectively communicating those insights to generate value. *Data engineering,* on the other hand, is an engineering domain that's dedicated to building and maintaining systems that overcome data processing bottlenecks and data handling problems for applications that consume, process, and store large volumes, varieties, and velocities of data. In both data science and data engineering, you commonly work with these three data varieties:

- **Structured:** Data that is stored, processed, and manipulated in a traditional *relational database management system (RDBMS)* – an example of this would be a MySQL database that uses a tabular schema of rows and columns, making it easier to identify specific values within data that's stored within the database.
- **Unstructured:** Data that is commonly generated from human activities and doesn't fit into a structured database format. Examples of unstructured data is data that comprises email documents, Word documents or audio / video files.
- **Semistructured:** Data that doesn't fit into a structured database system but is nonetheless organizable by tags that are useful for creating a form of order and hierarchy in the data. XML and JSON files are examples of data that comes in semi-structured form.

It used to be that only large tech companies with massive funding had the skills and computing resources required to implement data science methodologies to optimize and improve their business, but that's not been the case for quite a while now. The proliferation of data has

created a demand for insights, and this demand is embedded in many aspects of modern culture — from the Uber passenger who expects the driver to show up exactly at the time and location predicted by the Uber application to the online shopper who expects the Amazon platform to recommend the best product alternatives for comparing similar goods before making a purchase. Data and the need for data-informed insights are ubiquitous. Because organizations of all sizes are beginning to recognize that they're immersed in a sink-or-swim, data-driven, competitive environment, data know-how has emerged as a core and requisite function in almost every line of business.

What does this mean for the average knowledge worker? First, it means that everyday employees are increasingly expected to support a progressively advancing set of technological and data requirements. Why? Well, that's because almost all industries are reliant on data technologies and the insights they spur. Consequently, many people are in continuous need of upgrading their data skills, or else they face the real possibility of being replaced by a more data-savvy employee.

The good news is that upgrading data skills doesn't usually require people to go back to college, or — God forbid — earn a university degree in statistics, computer science, or data science. The bad news is that, even with professional training or self-teaching, it always takes extra work to stay industry-relevant and tech-savvy. In this respect, the data revolution isn't so different from any other change that has hit industry in the past. The fact is, in order to stay relevant, you need to take the time and effort to acquire the skills that keep you current. When you're learning how to do data science, you can take some courses, educate yourself using online resources, read books like this one, and attend events